Mini Project 3 – Country Comparison

Business Insight Report

Based on the five-page Country Comparison dashboard (Overview, Health & Education, Economic Analysis, Energy Analysis, Infrastructure Analysis).

1. Executive Summary

The dashboard compares roughly 250 countries and territories across demographics, health, economy, energy and infrastructure. Five messages stand out:

- **Economic power is highly concentrated.** The United States (11.75T), the European Union (11.65T), China (7.26T), Japan (3.75T) and India (3.32T) lead the GDP ranking, out of a global total of about 67.3T.
- **Income level predicts outcomes.** Higher-income regions sit at roughly 22K–36K on the regional comparison, while middle-income groups sit at about 6K–14K and lower-income groups at about 2K. The same pattern shows up in life expectancy, infant mortality and internet adoption.
- **The fastest-growing economies are mostly volatile, not stable.** Iraq (52.3%), Chad (38.0%), Liberia (21.8%) and Equatorial Guinea (20.0%) top the growth chart, and several of them also appear on the inflation and mortality charts.
- **Energy flows are uneven.** The EU consumes far more natural gas (467.7bn) than it produces (242.6bn), while Russia, Canada, Algeria and Norway are the main gas exporters. Saudi Arabia (7.9M) and Russia (6.1M) lead oil exports.
- **Several headline KPIs are distorted by data issues** (zero values for territories, the EU counted as a country). These should be fixed before the dashboard is used for decisions (see Section 4).

2. Key Findings by Theme

2.1 Global Overview

- Headline KPIs: average GDP 255.85bn, total population 7bn, highest GDP country United States, lowest inflation country Nauru.
- The top five economies by total GDP are the United States, European Union, China, Japan and India. The US and the EU are almost equal in size, and China is about 60% of either.
- Regional income gaps are large. Western Europe (36K), North America (32K) and Northern Europe (30K) lead among higher-income groups. In the same regions, the middle-income group is far lower (8K in North America, 13K in Northern Europe), which shows how unequal development is even inside one region.
- Lower-income groups are small and flat (about 2K) in almost every region, so regional location matters less than income class.

2.2 Health & Education

- Headline KPIs: average life expectancy 57.91, birth rate 18.95, death rate 8.02, infant mortality 30.25.
- Highest life expectancy: Andorra (83.51), Macau (82.03), San Marino and Singapore (81.62), Hong Kong (81.39), Japan (81.15). These are small, wealthy or highly urbanised economies.

- Highest birth rates (44–48) are in Niger, Uganda, Afghanistan, Mali, Chad, Somalia, Angola, Congo, Liberia, Burkina Faso and Malawi. Their death rates are also high (13–26), so population growth there is driven by fertility, not by good health outcomes.
- Infant mortality is lowest in Europe (5–6 per 1,000) and highest in parts of Africa. The Infrastructure page shows Nauru (90), Liberia (85), Zimbabwe (70) and Turkmenistan (60) at the top.
- The GDP vs. Life Expectancy scatter shows that life expectancy rises sharply with income at low GDP levels and then flattens. Extra GDP beyond a moderate level adds little, so targeted health spending in poorer countries has the best return.

2.3 Economic Performance

- Headline KPIs: average GDP growth 3.85%, average inflation 4.48%, exports 10T, unemployment 11.14%, average trade balance –49M.
- Highest growth: Iraq 52.3, Chad 38.0, Liberia 21.8, Equatorial Guinea 20.0, Venezuela 16.8. Lowest growth: Zimbabwe –8.2, Haiti –3.5, Saint Kitts and Nevis –1.9, Aruba –1.5, Burma –1.3.
- Growth outliers are mostly low-income or resource-dependent economies recovering from a very low base. The growth-vs-GDP-per-capita scatter confirms that lower-income points are spread widest on growth, while higher-income countries cluster at 0–5% growth.
- Highest inflation: the top country is at 55 (its label is cut off on the dashboard), followed by Angola (44), Iraq (25), Suriname (23), Venezuela (22) and Haiti (22). Iraq, Venezuela and Haiti appear on both a growth/decline list and the inflation list, which signals instability.
- Trade balance: the US imports about 1.5T against 0.8T of exports, a large deficit. Germany runs a surplus (0.9T exports vs. 0.7T imports). The EU, China, Japan and France are close to balance.

2.4 Energy & Natural Resources

- Headline KPIs: oil production 78.66M, gas production 2.88T, electricity generation 18.85T, 89 countries with oil reserves.
- Electricity: the US (3,839bn generated vs. 3,660bn consumed), the EU (2,888bn vs. 2,661bn), China (1,910bn vs. 1,630bn) and Japan (1,044bn vs. 971bn) all generate slightly more than they consume.
- Natural gas: the EU produces 242.6bn and consumes 467.7bn, so it relies on imports for about half of its demand. Algeria produces 80.3bn with almost no domestic consumption, which makes it an export-oriented supplier.
- Top gas exporters: Russia (0.17T), Canada (0.09T), EU (0.08T), Algeria (0.06T), Norway (0.05T). Top oil exporters: Saudi Arabia (7.9M), Russia (6.1M), EU (5.3M), Norway (3.5M).
- Reserve vs. production: most countries cluster near zero. One country has reserves near 47T and another near 27T, with production far below what the reserves could support, so reserves are largely unexploited or concentrated in a few countries.
- Russia and Norway appear on both export lists, so they are diversified energy suppliers and strategically important to Europe.

2.5 Infrastructure & Digital Access

- Headline KPIs: 908M internet users, 193M internet hosts, 33M km of highways, 1M km of railways.
- Internet adoption is highly concentrated. The EU has about 206M users and 22M hosts, and a handful of large markets account for most of the users on the scatter chart.

- Many low-income countries (for example Afghanistan with 1,000 users and Angola with 41,000) have almost no connectivity. Markets such as Argentina (4.1M users) and Australia (9.5M users, 2.8M hosts) show higher penetration relative to their size.
- Infrastructure and health are linked: countries at the top of the infant-mortality chart are also the ones with the weakest connectivity.

3. Cross-Cutting Insights

1. **Income is the strongest common driver.** GDP per capita lines up with life expectancy, infant mortality, internet access and energy use across all pages.
2. **Big markets and fast markets are different.** The largest economies grow at 0–5%, while the fastest growers are small, volatile and high-risk.
3. **Energy security is a strategic dividing line.** Producers (Russia, Saudi Arabia, Norway, Algeria, Canada) hold bargaining power over import-dependent regions such as the EU.
4. **Demographics split the world.** High-fertility, low-income countries (Niger, Uganda, Mali, Chad) will add population quickly, while wealthy countries have low birth rates and long lives.

4. Data Quality Issues to Fix

These issues change the conclusions, so they should be corrected before the dashboard is shared.

Issue	Where it appears	Impact	Suggested fix
Zero values for uninhabited areas and territories (Antarctica, Indian Ocean, Southern Ocean and similar)	Health & Education table	Pulls down averages: avg life expectancy of 57.91 is far below what the country list suggests; Northern Europe shows 65 and Oceania 38	Treat zeros as blanks and exclude non-country entities
European Union listed as a country	All pages	Double counts economies such as Germany, France and Italy, which also appear on their own; inflates the 67.3T total GDP	Separate the EU from individual countries, or add an "Entity Type" filter
Averages used for skewed measures	Overview and Economic KPIs	Avg GDP of 255.85bn is dominated by a few giants; avg unemployment (11.14) treats tiny and large countries equally	Use medians or population-weighted averages
Unlabeled units and years	Most charts	Readers cannot tell if values are per capita, PPP or which year	Add units and a data-year note on every page
Mislabeled or truncated visuals	Infrastructure scatter titled "GDP vs Life Expectancy" but showing internet users vs. GDP per capita; cut-off labels; "Execusive" and "Countreis" typos	Confusing for readers	Correct titles, widen labels, proof-read
Possible outliers	Nauru infant mortality 90; Iraq growth 52.3	May reflect small populations or unusual years rather than a real trend	Verify against the source and add a note

5. Recommendations

For decision-makers (use of the findings)

- **Prioritise markets by both size and stability.** Pair the top-GDP economies (US, EU, China, Japan, India) with a stability screen (inflation, growth volatility) before ranking fast-growth markets such as Iraq or Chad.
- **Treat high-inflation and negative-growth countries as high risk.** Venezuela, Haiti, Angola and Zimbabwe need hedging, short contracts or local partnerships.
- **Target health and infrastructure investment where the gap is widest.** Low-income countries with high birth rates and high infant mortality (for example Liberia, Niger, Mali and Chad) show the largest potential improvement per dollar.
- **Plan for energy dependence.** Import-dependent regions should diversify suppliers, and companies in energy-intensive sectors should factor in supply risk from a small group of exporters.
- **Look at digital growth in under-connected markets.** Very low internet access combined with large populations is a long-term opportunity for telecom, fintech and e-commerce players.